

Dragon Spellcasting System — Current Implementation Analysis

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Scope: D&D 3.5e Unity Prototype (/home/ubuntu/dnd35prototype , master branch)

1. Executive Summary

Dragon spellcasting is **partially implemented but entirely non-functional**. The data layer (DragonData.cs, NPCDatabase_Dragons.cs) has the scaffolding for sorcerer spell assignment, but two critical blockers prevent any dragon from ever casting a spell:

1. **IsSpellcaster always returns false** — Dragons use `CharacterClass = "Warrior"`, and `WarriorClass.IsSpellcaster` is `false`. The `SorcererCasterLevel` from `DragonAgeStats` is never propagated to the character's class system.
2. **SpellcastingComponent is never initialized** — The NPC setup code (`GameManager.NPCSetup.cs: 679`) gates spellcasting initialization on `stats.IsSpellcaster`, which is always false for dragons.

Even if these blockers were fixed, the spell lists are **severely limited** — only 4 of 10 dragon types have any spells defined at all, and those that do only know `mage_armor` and `shield` (two 1st-level defensive buffs) regardless of age or caster level.

2. Architecture — How Dragon Spells Are Currently Determined

2.1 Data Layer: DragonData.cs

Each `DragonTypeTemplate` has a single **flat spell list** shared across all age categories:

```
// DragonTypeTemplate (line 139)
public List<string> SorcererSpellIds; // Same for ALL ages of this dragon type
```

Each `DragonAgeStats` entry has a **per-age caster level**:

```
// DragonAgeStats (line 101)
public int SorcererCasterLevel; // 0 = no casting at this age
```

2.2 Assignment Layer: NPCDatabase_Dragons.cs

When a dragon is spawned (lines 98-108):

```
List<string> knownSpells = new List<string>();
List<string> preparedSlots = new List<string>();
if (stats.SorcererCasterLevel > 0 && template.SorcererSpellIds != null && template.SorcererSpellIds.Count > 0)
{
    knownSpells.AddRange(template.SorcererSpellIds);
    // "Simplified" - gives slots equal to min(casterLevel, spellCount)
    for (int s = 0; s < Mathf.Min(stats.SorcererCasterLevel, template.SorcererSpellIds.Count); s++)
        preparedSlots.Add(template.SorcererSpellIds[s]);
}
```

These are stored in the NPCDefinition:

```
KnownSpellIds = knownSpells, // line 169
PreparedSpellSlotIds = preparedSlots, // line 170
```

2.3 Initialization Blocker: GameManager.NPCSetup.cs

```
// Line 679 - THIS IS THE BLOCKER
bool shouldInitSpellcasting = stats.IsSpellcaster // ← Always FALSE for dragons!
    && ((def.KnownSpellIds != null && def.KnownSpellIds.Count > 0)
        || (def.PreparedSpellSlotIds != null && def.PreparedSpellSlotIds.Count > 0));

if (shouldInitSpellcasting)
{
    SpellcastingComponent spellComp = npc.Spellcasting
        ?? npc.gameObject.AddComponent<SpellcastingComponent>();
    // ... init spells ...
}
```

Because `CharacterClass = "Warrior"` and `WarriorClass.IsSpellcaster => false`, the `SpellcastingComponent` is **never created**.

2.4 AI Layer: AIService.ExecuteDragonTurn

```
// Line 648-656 - Would try casting, but IsSpellcaster is always false
if (!dragonProfile.IsTooCloseForCasting(npc, allCombatants, _gameManager))
{
    if (npc.Stats.IsSpellcaster && TryExecuteSpellcastAction(npc, target)) // ← never true
    {
        Debug.Log($"[AI][Dragon] {npc.Stats.CharacterName} cast a spell (priority 1/2).");
        yield return new WaitForSeconds(0.8f);
        yield break;
    }
}
```

And in `TryExecuteSpellcastAction` (line 2348):

```
if (!caster.Stats.IsSpellcaster)
    return false; // ← always exits here for dragons
```

3. Current Data: Spells and Caster Levels by Dragon Type

3.1 Dragon Type Spell Definitions

Dragon	SorcererSpellIds	Has Spells?
Red	mage_armor , shield	✓ (2)
Blue	mage_armor , shield	✓ (2)
Green	(empty list)	✗
Black	(empty list)	✗
White	(empty list)	✗
Gold	mage_armor , shield	✓ (2)
Silver	mage_armor , shield	✓ (2)
Bronze	(empty list)	✗
Copper	(empty list)	✗
Brass	(empty list)	✗

3.2 Sorcerer Caster Levels by Age Category

Dragon	Wyrmling	Very Young	Young	Juvenile	Young Adult	Adult
Red	0	0	1	3	5	7
Blue	0	0	1	3	5	7
Green	—	—	—	—	—	—
Black	—	—	—	—	—	—
White	—	—	—	—	—	—
Gold	0	1	3	5	7	9
Silver	0	0	1	3	5	7
Bronze	0	0	0	1	3	5
Copper	—	—	—	—	—	—
Brass	0	0	—	—	—	—

(— = SorcererCasterLevel field not defined in the age stats, defaults to 0)

3.3 Spell-Like Abilities

All 10 dragon types have `SpellLikeAbilityIds = new List<string>()` — **completely empty**.

4. D&D 3.5e Monster Manual Compliance

4.1 What the Rules Say (MM p.67-77)

Per D&D 3.5e, **all true dragons cast spells as sorcerers** starting from a specific age:

- **Chromatic dragons** gain sorcerer casting at Young (age 3) or later
- **Metallic dragons** generally gain it earlier (Very Young or Young)
- The caster level equals the dragon's listed caster level for that age
- Spells known/per day follow the Sorcerer table (PHB p.54) for that caster level
- **Spell selection is DM's choice**, but the MM gives suggested spell lists

4.2 Caster Level Discrepancies (per MM)

Dragon	MM Start Age	Impl Start Age	MM Adult CL	Impl Adult CL	Notes
Red	Young (3)	Young (3) ✓	7th	7 ✓	OK
Blue	Young (3)	Young (3) ✓	7th	7 ✓	OK
Green	Young (3)	Never ✗	5th	0 ✗	Missing entirely
Black	Young (3)	Never ✗	3rd	0 ✗	Missing entirely
White	Juvenile (4)	Never ✗	1st	0 ✗	Missing entirely
Gold	Very Young (2)	Very Young (2) ✓	11th	9 ⚠	CL too low for adult
Silver	Young (3)	Young (3) ✓	7th	7 ✓	OK
Bronze	Juvenile (4)	Juvenile (4) ✓	5th	5 ✓	OK
Copper	Young (3)	Never ✗	5th	0 ✗	Missing entirely
Brass	Juvenile (4)	Never ✗	3rd	0 ✗	Missing entirely

Summary: 5 of 10 dragon types are completely missing sorcerer casting. 1 has a slightly incorrect adult CL.

4.3 Missing Age Categories

The implementation only covers **6 of 12** D&D 3.5e age categories:

- ✓ Implemented: Wyrmling, Very Young, Young, Juvenile, Young Adult, Adult
- ✗ Missing: Mature Adult, Old, Very Old, Ancient, Wyrm, Great Wyrm

Higher ages have higher caster levels and access to more powerful spells.

4.4 Spell Selection Problems

Even for dragons that have `SorcererSpellIds` defined, the selection is woefully inadequate:

Issue	Description
Only 2 spells	All casting dragons know only <code>mage_armor</code> and <code>shield</code>
No offensive spells	No damage/control spells (fireball, lightning bolt, etc.)
Same spells regardless of age	A CL1 Young dragon and a CL7 Adult know identical spells
No scaling by caster level	CL7 sorcerer should know ~14 spells (0th-3rd level)
Flat list, not per-age	<code>SorcererSpellIds</code> is on the template, not per-age
No spell-like abilities	Many dragons gain SLAs at specific ages (e.g., Red: locate object at juvenile)

4.5 What a CL7 Sorcerer Should Know (Adult Red/Blue/Silver Dragon)

Per the PHB Sorcerer table (p.54), a 7th-level sorcerer knows:

- **Level 0:** 7 cantrips known, 6/day
- **Level 1:** 4 spells known, 6/day
- **Level 2:** 2 spells known, 5/day
- **Level 3:** 1 spell known, 3/day

Current implementation: 2 spells total (both level 1). That's **~14 spells short**.

5. Spell Database Availability

The SpellDatabase has **128 spells accessible to Sorcerers**, sufficient for proper dragon spell selection:

Spell Level	Count	Example Spells Available
0 (cantrip)	3	Dancing Lights, Daze, Touch of Fatigue
1	18	Mage Armor, Shield, Magic Missile, Grease, Sleep, Cause Fear, etc.
2	34	Mirror Image, Invisibility, Web, Resist Energy, Blur, etc.
3	31	Fireball, Lightning Bolt, Haste, Dispel Magic, Displacement, etc.
4	22	Wall of Fire, Greater Invisibility, Stoneskin, Dimension Door, etc.
5	9	Cone of Cold, Wall of Force, Hold Monster, Telekinesis, etc.
6	6	Chain Lightning, Disintegrate, Globe of Invulnerability, etc.
7	2	Spell Turning, Plane Shift
8	2	Protection from Spells, Sunburst
9	1	Wish

6. What Needs to Be Fixed (Priority Order)

6.1 Critical — Make Dragon Spellcasting Actually Work

Fix 1: IsSpellcaster bypass for dragons

- Either add "Sorcerer" to the dragon's class levels when `SorcererCasterLevel > 0`
- Or modify `GameManager.NPCSetup.cs` line 679 to also check if the `NPCDefinition` has a non-zero `SorcererCasterLevel` or non-empty `KnownSpellIds`
- The cleanest approach: Add a `SorcererLevel` field to `NPCDefinition`, and in `NPCSetup`, add a Sorcerer class level if it's > 0

Fix 2: Propagate caster level to CharacterStats

- Currently `SorcererCasterLevel` is never stored on the instantiated character
- The `SpellcastingComponent.Init()` needs a caster level to determine slots/day

6.2 High — Add Missing Dragon Caster Levels

Add `SorcererCasterLevel` to `DragonAgeStats` for the 5 missing dragon types:

- **Green:** Young=1, Juvenile=3, YoungAdult=5, Adult=5 (per MM p.75)
- **Black:** Young=1, Juvenile=1, YoungAdult=3, Adult=3 (per MM p.70)
- **White:** Juvenile=1, YoungAdult=1, Adult=1 (per MM p.78)
- **Copper:** Young=1, Juvenile=3, YoungAdult=5, Adult=5 (per MM p.83)
- **Brass:** Juvenile=1, YoungAdult=3, Adult=3 (per MM p.80)

6.3 High — Expand Spell Lists Per Age

Replace the flat `SorcererSpellIds` with per-age spell selection. Design options:

Option A: Per-age spell lists in `DragonAgeStats`

```
// Add to DragonAgeStats
public List<string> SorcererSpellsKnown; // Spells known at this age
```

Option B: Algorithmic selection based on CL

```
// Auto-generate spell lists from the sorcerer spells/known table
// using curated pools per dragon type (themed to their element)
```

Recommended thematic spell selections per dragon type:

Dragon	Theme	Suggested Spells
Red	Fire, destruction	burning_hands , fireball , wall_of_fire , fire_shield , haste
Blue	Lightning, illusion	shocking_grasp , lightning_bolt , mirror_image , blur , invisibility
Green	Poison, enchantment	charm_person , sleep , suggestion , stinking_cloud , dominate_person
Black	Acid, darkness	darkness , cause_fear , ray_of_enfeeblement , web , fear
White	Cold (minimal casting)	grease , shield (White dragons are the weakest casters)
Gold	Fire, protection	fireball , protection_from_energy , stoneskin , haste , dispel_magic
Silver	Cold, healing	shield , mirror_image , protection_from_energy , slow , haste
Bronze	Lightning, water	lightning_bolt , fog_cloud , resist_energy , displacement
Copper	Acid, trickery	grease , web , slow , confusion , stoneskin
Brass	Fire, talk	sleep , charm_person , hold_person , suggestion

6.4 Medium — Add Spell-Like Abilities

Many dragons gain specific spell-like abilities at certain ages. These are separate from their sorcerer spellcasting:

Dragon	SLA	Age Gained
Red	Locate Object	Juvenile
Blue	Create/Destroy Water (at will)	Wyrmling
Green	Suggestion	Juvenile
Black	Darkness (at will, 3×/day)	Juvenile
White	Fog Cloud	Young
Gold	Bless	Very Young
Silver	Fog Cloud	Very Young
Bronze	Speak with Animals (at will)	Wyrmling
Copper	Spider Climb (at will)	Wyrmling
Brass	Speak with Animals (at will)	Wyrmling

6.5 Low — Sorcerer Slot System

The current “simplified” slot assignment (`min(casterLevel, spellCount)`) doesn’t match the sorcerer spells-per-day table. A proper implementation would need:

- Spells known by level (from PHB sorcerer table)
- Spells per day by level (from PHB sorcerer table)
- Spontaneous casting (can cast any known spell using available slots)

7. Quick Fix vs. Full Fix

Quick Fix (minimal changes to make existing spells work)

1. In `GameManager.NPCSetup.cs` line 679, change condition to:

```
csharp
bool shouldInitSpellcasting = (stats.IsSpellcaster || (def.KnownSpellIds != null &&
def.KnownSpellIds.Count > 0))
&& ((def.KnownSpellIds != null && def.KnownSpellIds.Count > 0)
|| (def.PreparedSpellSlotIds != null && def.PreparedSpellSlotIds.Count > 0));
```

This lets creatures with explicit spell lists (like dragons) bypass the class check.

2. Add `SorcererCasterLevel` to `NPCDefinition` and pass it through to `CharacterStats` so the spellcasting system knows the caster level.

Full Fix (D&D 3.5e compliant)

1. All items from Quick Fix
2. Add caster levels for all 10 dragon types (section 6.2)
3. Create proper per-age spell lists (section 6.3)

4. Add spell-like abilities (section 6.4)
5. Implement sorcerer slot system (section 6.5)
6. Consider adding the missing 6 age categories (Mature Adult through Great Wyrms)

8. Files Involved

File	Role	Status
Assets/Scripts/Character/Creatures/DragonData.cs	Dragon type templates & age stats	Has CL data for 5/10 dragons, flat spell lists
Assets/Scripts/Character/Creatures/NPCData-base_Dragons.cs	Builds NPCDefinition from template	Has spell assignment code, but never executes effectively
Assets/Scripts/_Core/GameManager.NPCSetup.cs	Initializes SpellcastingComponent	BLOCKER — gates on <code>IsSpellcaster</code> which is false for dragons
Assets/Scripts/Character/Stats/CharacterStats.cs	<code>IsSpellcaster</code> property	Checks class system, not NPC-defined caster levels
Assets/Scripts/Character/Classes/NPC/WarriorClass.cs	Warrior class definition	<code>IsSpellcaster => false</code>
Assets/Scripts/Services/AIService.cs	Dragon AI turn execution	Has spellcasting priority, but blocked by <code>IsSpellcaster</code>
Assets/Scripts/AI/Profiles/DragonAIProfile.cs	Dragon AI profile	Properly prioritizes spells before breath weapon
Assets/Scripts/Spell/Data-base/SpellDatabase_*.cs	Spell definitions	128 Sorcerer-accessible spells available
Assets/Scripts/Spell/Components/SpellcastingComponent.cs	Spell management	Would work if initialized